

CSS BOX-SIZING

Complete Explanation with Examples

1. What is box-sizing?

The CSS box-sizing property controls how the total width and height of an element are calculated.

In CSS, an element can have content, padding, and a border. The box-sizing property decides whether the width and height you specify include padding and border or only refer to the content area.

The two important values are: content-box and border-box.

2. content-box — Default Value

content-box is the default CSS box-sizing value. When you write width: 200px, the 200px refers only to the content area. Padding and border are added outside that width.

```
box-sizing: content-box;  
width: 200px;  
padding: 20px;  
border: 5px solid black;
```

Total width = content width + left/right padding + left/right border

Total width = 200 + 20 + 20 + 5 + 5 = 250px

3. border-box

With border-box, the width and height you specify include both padding and border. This makes sizing elements easier because the final outer size remains the specified size.

```
box-sizing: border-box;  
width: 200px;  
padding: 20px;  
border: 5px solid black;
```

Total width = 200px

The 200px includes the content area, 40px of horizontal padding, and 10px of horizontal border. Therefore, the content area becomes smaller so that the complete element remains 200px wide.

4. Visual Comparison

content-box

Width: 200px
+ Padding: 40px
+ Border: 10px

border-box

Width: 200px
Includes padding: 40px
Includes border: 10px

= Total: 250px

= Total: 200px

5. Commonly Used Code

A common practice is to apply border-box to all elements using the universal selector (*). This makes width and height calculations more predictable.

```
* {  
  box-sizing: border-box;  
}
```

This is why you often see `* { box-sizing: border-box; }` at the beginning of CSS files.

6. Why did we use it in Lab 8?

In the Lab 8 Part A layout, we used:

```
* {  
  box-sizing: border-box;  
}
```

This makes the declared width and height include padding and borders. It helps keep the layout dimensions predictable when padding is added to elements such as `.page`, `.header`, `.content`, `.entry`, and `.sidebar`.

7. Simple Example

```
.box {  
  width: 300px;  
  height: 150px;  
  padding: 20px;  
  border: 5px solid black;  
  box-sizing: border-box;  
}
```

Because border-box is used, the outside dimensions of `.box` remain exactly 300px wide and 150px high.

8. Quick Difference

Feature	content-box	border-box
Default?	Yes	No
Width includes padding?	No	Yes
Width includes border?	No	Yes
Easy for fixed layouts?	Less convenient	More convenient

Remember: content-box adds padding and border outside the declared size; border-box keeps padding and border inside the declared size.